PT100

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Day 1[5 October,Monday]

We got the components from the core labs and they initially gave us breadboard. So, we continued using them and learnt how to control the lcd.

Day 2[6 October,Tuesday]

We figured out how to control the lcd and we assembled the simple voltage divider circuit on the breadboard. We initially calculated the temperature using standard coefficients and the readings were all over the place.

Day 3[7 October,Wednesday]

We realized that we need to take the average of the readings and also improve the coefficients. The concept of soldering was informed

Day 4 [9 October,Friday]

We decided to start soldering the lcd and were not aware of the concept of using header pins. We soldered all pins of the lcd into the pcb.

Day 5[12 October,Monday]

We understood that our soldering guns were not proper and that we needed to change them.

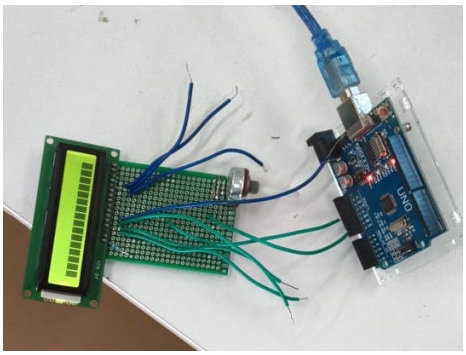


Figure 1: Your caption here

Day 6[13 October,Tuesday]

After receiving our improved guns, we soldered the data wires and the power circuitry to the lcd. When we started the lcd, only half of the displays were printing.

Day 7[15 October,Thursday]

We spent hours checking all the pin voltages and determined that they were all proper. We decided to remove and change the lcd and the pcb. We also got the header pins and carefully soldered them to the pcb, avoiding shorting any pins.

When we inserted the second lcd, it lighted up but did not display anything, even when the pin voltages were all correct. We got another teams lcd and it worked perfectly, so we understood that the second lcd was also defective. We finally soldered the rest of the simple voltage divider circuit while also simultaneously developing the 3d model.

The code that was taken from the github repository used the voltage difference between the resistor and 2.5v, which usually comes to around 0.1v(for room temperature) , and eliminates noise. Since we used a simple voltage divider, the code was adjusted accordingly and we realized that the error of the analogRead() , subtracted with 2.5, was always coming to zero, forcing us to abandon that code.

We developed 2 codes. One takes the average and finds the temperature. The other calculates the exact resistance of the constant resistor using boiling water and it came out to be around 120 ohm.(We assumed the coefficients of the internet to be valid.) .

Due to the time constraints (caused by the repeated failure of the lcds), we have

decided not to build an innovated model,so we had a simple 3D model of box with a cap with a suitable holes. As it was easy to do.

1 Day 8[15 October,Saturday]

Using the lab thermometer readings and voltage output from display.

We calibrated the values of a,b,c using LSQ. The eqation is:

$$T = aV^2 + bV + c$$

where a,b,c are the constants we need to determine. Quadratic least square Fit graph:

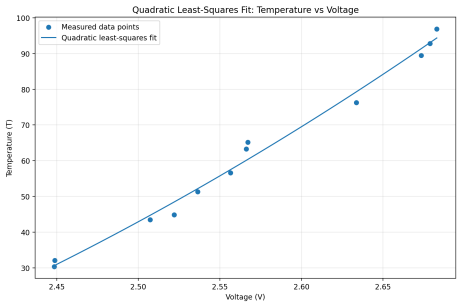


Figure 2: Your caption here

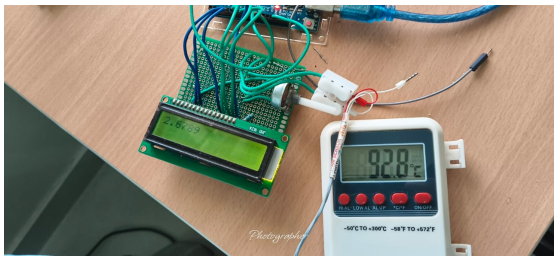


Figure 3: Your caption here

The results

The temperature output was showing around room temperature but after waiting for a while, it keeps oscillating between 27.95 to 29.05. The R^2 value was 0.98782 and we took 12 readings.